EXPERIMENT – 24 : Prolog Program to Implement a Diet Recommendation System

Aim:

To write and execute a Prolog program to recommend a suitable diet based on a person's health condition and food preferences using facts, rules and queries.

Program

File Name: exp24.pl

File Path: C:/ai prolog/exp24.pl

```
% Diet Recommendation Syste

% Facts - Health Conditions

health_condition(diabetes).

health_condition(obesity).

health_condition(underweight).

health_condition(normal).

% Diet Recommendations

diet(diabetes, low_sugar).

diet(diabetes, whole_grains).

diet(diabetes, green_vegetables).

diet(obesity, low_calorie).

diet(obesity, fruits).

diet(obesity, green_vegetables).

diet(underweight, high_protein).

diet(underweight, healthy_fats).

diet(underweight, milk).

diet(normal, balanced_diet).

diet(normal, fruits).

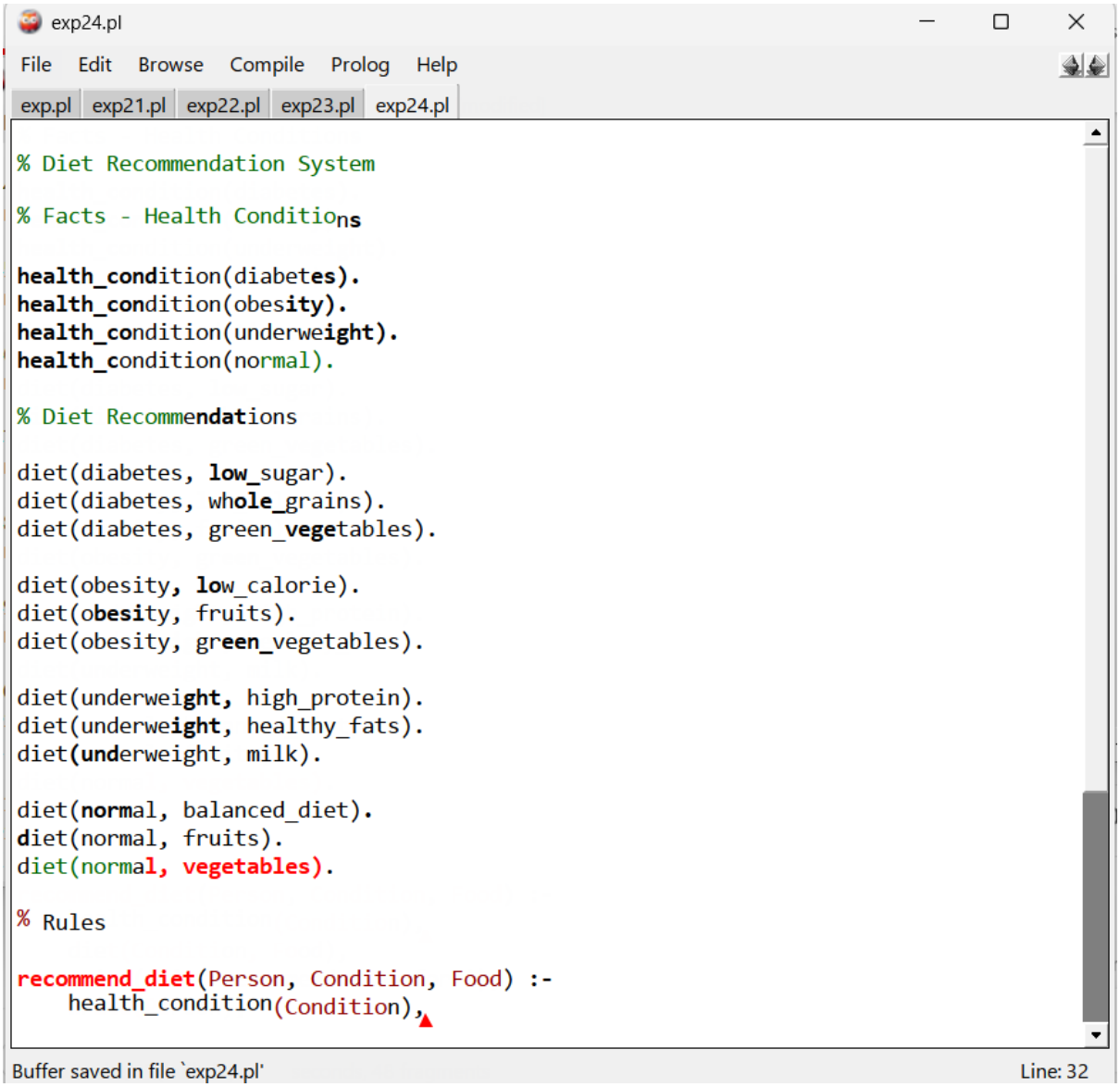
diet(normal, vegetables).

% Rules


recommend_diet(Person, Condition, Food) :-
```

```
    health_condition(Condition),
    diet(Condition, Food),
    write('Diet recommendation for '),
    write(Person),
    write(' is: '),
    write(Food).
% Invalid Condition

recommend_diet(Person, Condition, _) :-
    \+ health_condition(Condition),
    write('Invalid health condition for '),
    write(Person).
```



Procedure

1. Open SWI-Prolog software.
2. Create a new file and enter the above Prolog program.
3. Save the file as exp24.pl in C:/ai prolog/.
4. Load the program using the following command:

?- consult('C:/ai prolog/exp24.pl').

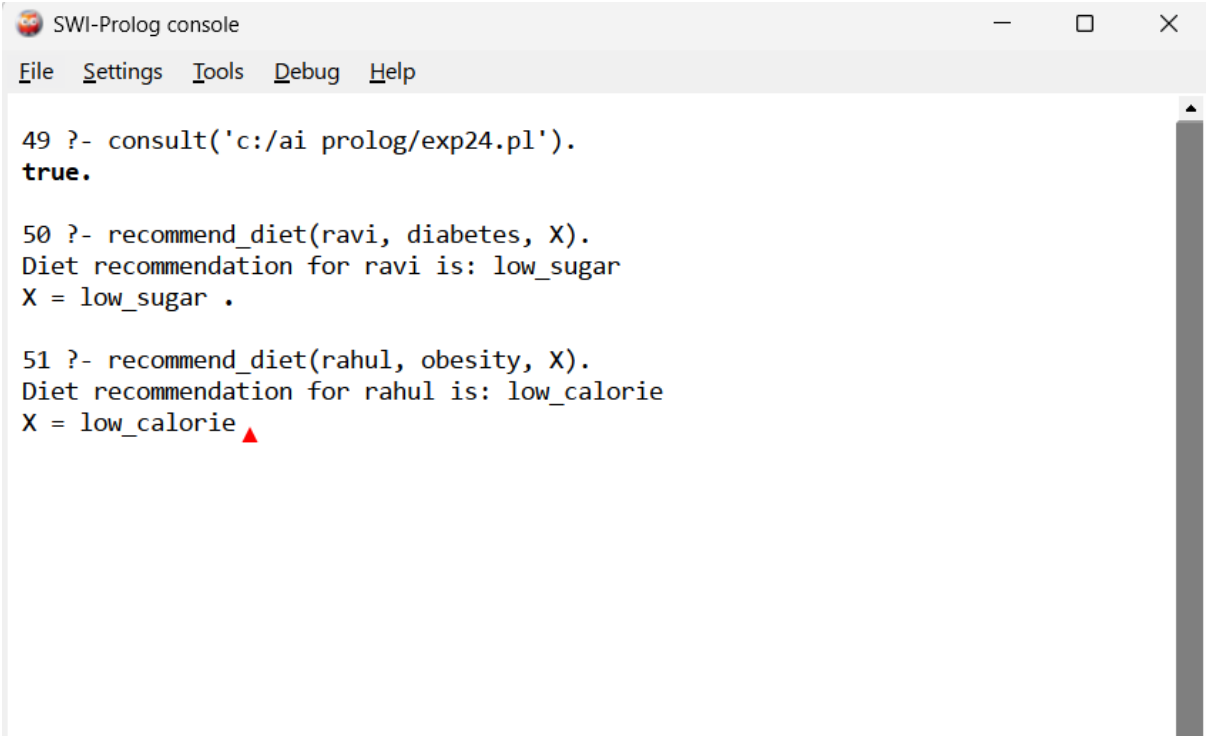
5. Execute the required queries to obtain diet recommendations.
6. Observe the output in the Prolog console.

Required Queries and Output

Query 1: Diet recommendation for a person with diabetes.

?- recommend_diet(ravi, diabetes, X).

Output:



Explanation

1. Facts: Define different health conditions.
2. Diet: Stores the recommended food categories for each health condition.
3. Rules: Match a person's health condition with suitable diet recommendations.
4. Queries: Ask Prolog to display the recommended diet.

Result

Thus, the Prolog program to implement a diet recommendation system using facts, rules and queries was written and executed successfully.